BUIDL CTC 2026 FALL · DEFI

Kitty

Savings circles where every payment is proven, not promised.

Ethereum stablecoins · settled and credit-scored on Creditcoin · via the Attestcoin Protocol

PROBLEM

Hundreds of millions save in circles nobody can see.

- Chit funds, susu, tandas, chamas, stokvels: ten friends, one pot a month.
- They fail two ways: the treasurer runs, or a member stops paying and nobody outside the group ever knows.
- Ten years of perfect payments builds zero formal credit history.

TRUST TODAY

Who you have to trust

STEP	INFORMAL CIRCLE	“ROSCA ON-CHAIN” APPS	KITTY
Holds the money	Treasurer	One contract on one chain	Escrow vault on Ethereum
Knows who paid	Treasurer’s notebook	Same contract	Attestcoin-proven txs on Creditcoin
Decides the deadline	Treasurer	block.timestamp / admin	Attested source-chain block height
Picks the recipient	Treasurer / lottery	VRF oracle	Fixed order or by proven score
Credit history	None	None	Kitty Score, readable by any lender

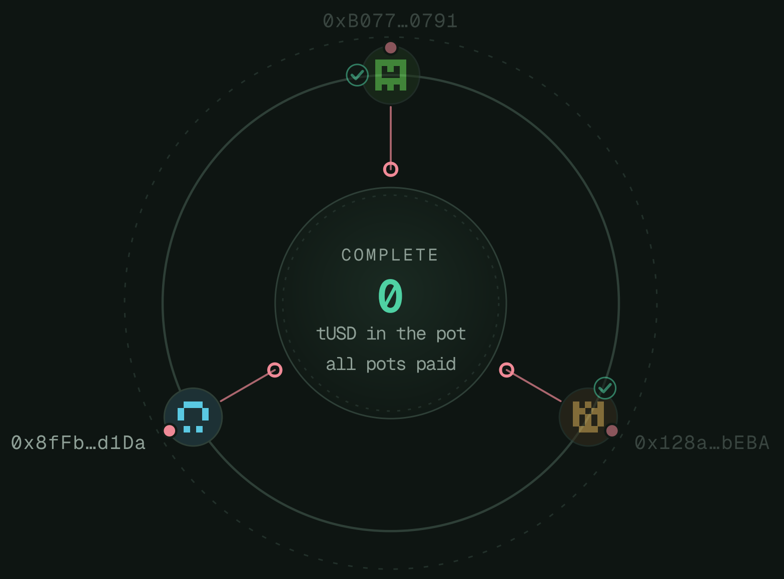
SOLUTION

Money on Ethereum. Rules on Creditcoin. Proofs in between.

- Members pay into a minimal escrow vault on Sepolia.
- The attestor network attests the block; one batch proof covers the whole round.
- KittyLedger verifies through precompile 0x0FD2, decodes and binds every payment, enforces deadlines from 0x0FD3.
- Payouts are proven back. “Paid” is never an operator’s word.
- Any member can prove a round from the browser: the Proof Builder serves CORS, the ledger doesn’t care who submits.

LIVE LOOP · PAY → ATTEST → PROVE → VERIFY → CLOSE → PAY OUT → PROVE BACK

The first circle, live: Delhi Chit Circle



● proven on time ● proven late ● paid · proof pending ● missed

— receives this round ✓ already received

3 members · 100 tUSD a round · rotation by Kitty Score · round 3 of 3

CIRCLES
3
live · Creditcoin CC3 Testnet

PAYMENTS PROVEN
26
92% on time

BATCH PROOFS
8
26 tx through 0x0FD2

TUSD SETTLED
1,600
payouts proven back

ATTESTED SEPOLIA BLOCK
11688250
latest · 0x0FD3

KITTYLEDGER
0xC2A1…F276
Creditcoin CC3 Testnet · Blockscout

ATTESTCOIN DEPTH

What the ledger actually checks

- Batch `verifyAndEmit`: up to 10 txs, one continuity proof, one call for payments pooled across every open circle, preflighted free with the view verify, 24% cheaper than singles on the live 0x0FD2.
- EvmV1Decoder: receipt status 1, exactly one Contributed log, emitter = registered vault, tx.to = vault, tx.from = member, exact amount, current round.
- Query id = keccak(chainKey || height || txIndex), same as ASCBase; per-circle chain key validated with `get_chain_by_key`; vault allowlist keyed by chain.
- Deadlines from attested height: `is_height_attested(chainKey, deadline + 64)`; a miss records the attestation that proved it.
- The dashboard asks 0x0FD3 which attestation covers a payment: `find_lowest_attested_after`, `get_attestation_bounds`.
- Proven against the live precompile before deploying: a real Sepolia proof → `0x0FD2.verify = true`; tampered bytes and a wrong chain key revert (`pnpm verify:live`).

src/asc/KittyLedger.sol · recordContributions

one precompile call

```
function recordContributions(
    uint64 chainKey,
    uint64[] calldata heights,
    bytes[] calldata encodedTxs,
    INativeQueryVerifier.MerkleProof[] calldata merkleProofs,
    INativeQueryVerifier.ContinuityProof calldata continuity
) external {
    bytes32[] memory queryIds = _prepareBatch(chainKey, heights, encodedTxs, merkleProofs);
    uint256 n = heights.length;

    bool ok = VERIFIER.verifyAndEmit(chainKey, heights, encodedTxs, merkleProofs, continuity);
    if (!ok) revert ProofRejected();

    uint64 lo = type(uint64).max;
    uint64 hi;
    for (uint256 i; i < n; ++i) {
        processedQueries[queryIds[i]] = true;
        _recordContribution(chainKey, queryIds[i], heights[i], encodedTxs[i]);
        if (heights[i] < lo) lo = heights[i];
        if (heights[i] > hi) hi = heights[i];
    }
    emit BatchVerified(chainKey, lo, hi, n);
}
```

ATTACK LAB

Eight ways to cheat: six decoded rejections, two accepted by design

- Replay → QueryAlreadyProcessed
- Fake vault with a perfect event → WrongEmitter
- Same proof, chain key 3 → WrongChain
- Included but reverted tx → SourceTxFailed
- Paid after the deadline → accepted, flagged late, −20
- Steal the steward's key → NotOperator / OwnableUnauthorizedAccount / RoundStillOpenOnSource / VaultNotTrusted
- Fire the agent → accepted: the ledger checks the proof, not the caller
- Poison the reasoning → three invented figures stripped before display

Reviewed adversarially before submission: only owner-trusted vaults can feed the ledger, nobody can be penalised for a circle they never consented to, a 64-block grace window protects on-time payers from a hostile early close, and pots only go to members who paid this round.

KITTY STEWARD

An agent whose only power is proof

LAYER	HOLDS	WHAT IT DOES
1 · The ledger	final say	Verified proof, receipt status 1, trusted emitter, right sender and target, exact amount, current round, unseen query id. Any failure: nothing happens.
2 · Deterministic decisions	timing only	Prove now or wait for a fuller batch? Missing the grace window costs 120 points: urgency beats thrift. Decisions logged.
3 · Cited reasoning	none	Claude explains the log in plain language. Any sentence with an uncited or unverifiable figure is stripped before display. No API key: the deterministic sentence, identical behaviour.

THE STEWARD'S LAST THREE DECISIONS · RECORDED ON CC3 TESTNET

web/src/data/steward.sample.json · 24 entries

prove 12 Sept, 10:13 UTC

proving 1 payment(s) from 1 circle(s) on chain key 1 in one call: waited 478s for a fuller batch

chainKey=1 queries=1 circles=1 attestedHeight=11688220 sourceHead=11688252 blocksOfSlack=45 roundsCompleted=0 blockSpan=0 leftBehind=0 waitedSeconds=478 stillWaitingForAttestation=0 preflight=passed queriesInCall=1 fromHeight=11688211 toHeight=11688211 continuityRoots=10

creditcoin 0x43028c3850… sepolia 0x0f83e10d09…

close 12 Sept, 10:13 UTC

closed circle 2 round 2 — every member proven

circleId=2 round=2 closeHeight=11688297 proven=5 members=5 everyoneProven=true

creditcoin 0xc1eca80ff1…

payout 12 Sept, 10:13 UTC

paid circle 2 round 2: 250 tUSD to 0x8fFb6727EaE2F3C5Ef9F68Ac0ac2dA0caEc2d1Da

circleId=2 round=2 recipient=0x8fFb6727…d1Da amount_tUSD=250

sepolia 0x8425800f16…

KITTY SCORE → CREDIT

The score is used, not just displayed

- 500 base · +15 on time · −20 late · −120 missed · 300–850.
- KittyCreditLine on Creditcoin underwrites from the score alone: tier A borrows 100% of proven volume, B 50%, C 20%, D nothing.
- A soulbound Kitty Score badge renders live from the ledger, so a lender can read it on any explorer.
- Every input is a proven transaction or an attested deadline. This is the data Creditcoin was built to carry.

LIVE ON TESTNET

The first circle has already settled

- KittyLedger, KittyViewer, KittyUSD, KittyCreditLine and KittyBadge deployed to Creditcoin CC3 Testnet on 12 September 2026 (KittyLedger 0xC2A1…F276); its paired Sepolia vault 0xa27e…DA84 is trusted on the ledger; the first circle and all eight attack scenarios ran on the first ledger and vault.
- Delhi Chit Circle, three members, 100 tUSD a round, rotation by score: three Sepolia payments, one batch proof of three verified by the live 0x0FD2 in a single call, an early close, a 300 tUSD payout on Sepolia and its proof back to Creditcoin. Then two more circles: eight payments settled in one cross-circle call, a member proving from the browser, and a real miss closed on the attested deadline.
- Attack scenarios rerun against the real precompile: the forged chain key is rejected by 0x0FD2 itself. Every transaction is in docs/TESTNET_LOG.md; the hosted dashboard, steward log and attack lab read the same chain.

ACTION	CHAIN	TRANSACTION	GAS
deploy KittyLedger · the final ledger, 0xC2A1…F276	Creditcoin	0x21994599…7ac7f5	4,901,416
recordContributions round 0 · the whole round, batch of 3, one 0x0FD2 call	Creditcoin	0xac2a637f…652fe9	888,573
closeRound 0 · everyone paid, early close, rotation by score	Creditcoin	0x53cbb51d…277bbb	354,326
payout round 0 · 300 tUSD to member 0	Sepolia	0xfee30618…7263c3	64,821
confirmPayout round 0 · proven back through 0x0FD2	Creditcoin	0x92344d88…74ba88	386,582
recordContributions · 8 payments from circles 2 and 3 in ONE precompile call	Creditcoin	0xa7310f00…c1f7fa	1,936,724
recordContributions circle 3 round 1 · proven from the browser by the member, no operator	Creditcoin	0x7b8fdaab…8861b5	401,366
closeRound circle 1 round 1 · attested deadline + 64, a real miss with its attestation	Creditcoin	0xef146316…9f01ca	362,992

ROADMAP · CEIP

From circles to credit lines

- Score-gated circle sizes and seat bidding.
- Lender integrations on Creditcoin.
- Attestcoin writability for payouts once audited; Ethereum mainnet chainKey on CC3 mainnet.
- Pot cover through proven-event insurance.

TEAM

Prashant · solo builder

Foundry · TypeScript · React. Repo: https://github.com/Prashant-thakur77/Kitty. Contracts on Sepolia and Creditcoin CC3 Testnet; 162 Foundry tests · 29 agent tests · 8 attack scenarios end to end in CI; verified against the live 0x0FD2.

0xB077…0791 member 0 · first pot received KittyLedger 0xC2A1…F276